

# Soumya Ramchandran

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## SKILLS

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**Data Analysis and Machine Learning** : Scikit-learn, Pandas, NumPy, TensorFlow, Matplotlib, Seaborn

**Programming** : Python, Java, C, JavaScript

**Development** : HTML, CSS, React

**Databases** : MySQL, PostgreSQL, MongoDB

**Tools** : GitHub, VS Code, Power BI, Excel

## PROJECTS

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### Black Friday Sales Analysis & Prediction System | Python, Pandas, NumPy, Matplotlib, Streamlit

- Built an end-to-end analytics and prediction system using **Python, Streamlit**, and **scikit-learn**.
- Implemented **Random Forest regression** to predict purchase amount based on customer and product features.
- Designed an **interactive dashboard** with filters, KPIs, and dynamic visualizations.
- Integrated personalized insights by comparing predictions with average purchase behavior.
- Deployed the application on **Streamlit Cloud** for real-time accessibility.

### Dynamic Multimodal Deep Learning Model for Lung-Cancer Survival Prediction | Python, PyTorch (Ongoing)

- Built RMSurv, a **trimodal deep learning model** fusing **gene expression, clinical, and CT imaging data** for lung cancer survival prediction, achieving **C-index of 0.709** on TCGA-LUAD
- **Validated externally on TCGA-LUSC** (unseen cohort, different subtype) without retraining, yielding **C-index 0.703 and Hazard Ratio 1.86** (High vs Low risk), confirming cross-subtype generalizability.
- Conducted full ablation study across 5 modality combinations, proving CT imaging as the key driver of external robustness. trimodal model improved **LUSC C-index by +9.7%** over bimodal (genomic + clinical)
- Engineered end-to-end pipeline in PyTorch + Lifelines with **Cox loss, 5-fold CV, bootstrap 95% CI, Kaplan-Meier curves, and log-rank testing**.

### Student Management System | Java, JDBC, PostgreSQL

- Built a backend system supporting **CRUD operations** for student records.
- Implemented **secure authentication and prepared statements** to prevent SQL injection.
- Designed **relational database schema** and integrated it with Java using JDBC.

## EDUCATION

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### Vellore Institute of Technology (Chennai Campus)

*Master of Computer Applications (MCA)*

July 2025 – July 2027

Chennai, Tamil Nadu

- **CGPA:** 8.69

### Maharashtra Institute of Technology

*Bachelor of Computer Applications (BCA)*

Aug 2022 – May 2025

Pune, Maharashtra

- **CGPA:** 8.46

## CERTIFICATES

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### Deloitte Data Analytics Job Simulation

*Data Analytics Virtual Experience*

Apr 2025

Forage

- **Performed real-world analytics tasks including** data analysis and forensic technology case investigation.